

Proportions :

$$H_0: p = p_0$$

$$H_1: p \neq p_0$$

OR $p > p_0$

$$P(X \geq x_0 | p = p_0)$$

OR $p < p_0$

$$P(X \leq x_0 | p = p_0)$$

$$P\text{-value} = \begin{cases} 2P(X \leq x | p = p_0) & \text{if } x < np_0 \\ 2P(X \geq x | p = p_0) & \text{if } x > np_0 \end{cases}$$

Testing a
Proportion
(Small Samples)

1. $H_0: p = p_0$.
2. One of the alternatives $H_1: p < p_0, p > p_0$, or $p \neq p_0$.
3. Choose a level of significance equal to α .
4. Test statistic: Binomial variable X with $p = p_0$.
5. Computations: Find x , the number of successes, and compute the appropriate P -value.
6. Decision: Draw appropriate conclusions based on the P -value.

n is large, normal approximation with $\mu = np_0$ and $\sigma^2 = np_0q_0$ — from CLT
Under H_0 very accurate as long as p_0 is not ≈ 0

$$Z = \frac{x - np_0}{\sqrt{np_0q_0}} = \frac{\hat{p} - p_0}{\sqrt{p_0q_0/n}}$$

For two samples —

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1q_1}{n_1} + \frac{p_2q_2}{n_2}}}$$

Under $H_0: p_1 = p_2 = p$, implies

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$p \Rightarrow \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

For single population with variance.

$$\chi^2 = \frac{(n-1) s^2}{\sigma_0^2}$$

$$\chi_{n-1}^2$$

for test

$$H_0: \sigma^2 = \sigma_0^2 \quad \text{vs.}$$

$$H_1: \sigma^2 > \sigma_0^2 \quad \text{or} \quad \sigma^2 < \sigma_0^2 \quad \text{or} \quad \sigma^2 \neq \sigma_0^2$$

For two populations with variances

$$f = \frac{s_1^2}{s_2^2}$$

$$\sim F_{n_1-1, n_2-1} \quad \text{under } H_0$$

$$\text{for test } H_0: \sigma_1^2 = \sigma_2^2 \quad \text{vs.}$$

$$H_1: \sigma_1^2 > \sigma_2^2 \quad \text{or} \quad \sigma_1^2 < \sigma_2^2 \quad \text{or} \quad \sigma_1^2 \neq \sigma_2^2$$